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Abstract

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First-principles prediction of the thermoelectric properties of $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ alloys

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Thermoelectric power is an efficient and flexible technology for improving energy efficiency, but widespread applications require cost-effective and sustainable materials with a high figure of merit zT . Alloying is a common strategy for optimising zT , but predicting and understanding its impact on the material properties is made challenging by the structural disorder. In this work, we develop a practical and rigorous approach to predicting the zT of alloys by combining electrical-transport and lattice thermal-conductivity calculations, with approximate models for the electron and phonon scattering, and apply it to study the $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ alloy. We compare predictions obtained by averaging over the full set of unique configurations in a 32-atom supercell with those obtained from special quasi-random structure (SQS) structures with supercell sizes of up to 256 atoms. We find that a 128-atom SQS model achieves a good balance between computational efficiency and realistic predictions of the physical properties. The calculations indicate that $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ shows comparable performance to the SnSe endpoint but with a modest enhancement to the zT due to a reduction in the lattice thermal conductivity, which has a particularly significant impact on the low-temperature performance.

1 Introduction

60-70% of global energy is currently lost as heat, and technologies to improve energy efficiency and reduce global demand for fossil fuels are crucial enablers for the transition to net zero greenhouse gas emissions.¹⁻³ Thermoelectric (TE) power, which enables the direct recovery of waste heat to electricity, is a promising technology to meet this need, and its demonstrated applications include energy-intensive processes that are particularly challenging to decarbonise, such as heat recovery from automobile powertrains and industrial data centre facilities.^{4,5}

Thermoelectric generators (TEGs) leverage the Seebeck effect in a matched pair of TE materials with dominant p-type (hole) and n-type (electron) transport, connected electrically in series and thermally in parallel, to extract energy from a temperature gradient. The performance of a TE material is typically described by the dimensionless figure of merit zT given by:⁶

$$zT = \frac{S^2 \sigma T}{\kappa_{\text{latt}} + \kappa_{\text{el}}} \quad (1)$$

where S is the Seebeck coefficient, σ is the electrical conductivity,

$S^2 \sigma$ is the power factor (PF), and κ_{latt} and κ_{el} are the lattice (phonon) and electronic contributions to the total thermal conductivity κ . The electrical transport properties S , σ and κ_{el} depend on the electronic structure and can be balanced through the charge-carrier concentration n .⁷ The optimum balance between minimising κ_{el} and maximising the PF is usually found in heavily-doped semiconductors with $n \approx 10^{19}$ – 10^{21} cm^{-3} .⁷ The lattice thermal conductivity is generally considered to be independent of the electronic structure, and low κ_{latt} is associated with structures composed of heavy atoms, with weak and/or inhomogeneous chemical bonding, and with strong anharmonic phonon scattering.^{8,9} The κ_{latt} can also be modulated by atomic-scale point defects and by nano- and microstructuring, with the required length scales dependent on the characteristic wavelengths and mean-free paths of the heat-carrying phonon modes.¹⁰

Tin monoselenide (SnSe) has received significant attention as a promising TE material due to its favourable electronic structure and intrinsically low κ_{latt} ,^{8,11-13} and the high elemental abundance and relatively low toxicity of Sn and Se. Despite achieving record $zT \approx 2.6$ in bulk and $zT > 3$ in polycrystalline powders,^{14,15} the best performance is only obtained at high temperature, and the low-temperature performance is not competitive with the industry-standard Bi_2Te_3 (zT up to ~ 1.5 from 300–400 K^{16,17}). Given that the majority of waste heat is released below 100 °C,^{6,18,19} improving the low-temperature performance is critical for widespread application.

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Alloying to form solid solutions is an established technique for improving the performance of TE materials.^{20–26} Alloying can reduce the thermal conductivity,^{20–23} modulate the electronic band structure to improve the power factor,^{24,25} and can also improve chemical compatibility with dopants (“dopability”).²⁶ SnS and SnSe crystallise into isostructural orthorhombic phases (*Pnma* spacegroup) under ambient conditions and transform into a higher-symmetry *Cmcm* phase at high temperature.²⁷ The second-order phase transition results in both systems displaying similar anharmonic lattice dynamics.^{8,13,28} The two materials also have similar bandgaps and dielectric properties.^{29–32} This similarity makes the formation of $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ solid solutions facile, and various experimental studies have indicated that the alloys can show lower lattice thermal conductivity and improved zT compared to both the SnS and SnSe endpoints.^{33–35}

Advances in theoretical techniques, software and computing power over the past decade have made it possible to accurately and efficiently predict all four properties in Eq. 1, and the zT , using first-principles modelling techniques such as density-functional theory (DFT).^{36–41} We have developed an approach that combines models for the structural dynamics and electrical transport, including descriptions of the phonon and electron scattering, and successfully benchmarked it against *Pnma* SnS and SnSe, the related Ge chalcogenides, and, very recently, the novel “ π ” phases of the Sn monochalcogenides.^{42–44} More generally, modelling has also played a significant role in understanding the structure-property relationships that underpin high thermoelectric performance, paving the way toward the theory-led design of new materials.^{9,44–49} However, while theoretical methodology for studying bulk materials is well established, solid solutions and heavily-doped materials remain a significant challenge. We have previously predicted the bandgaps and dielectric properties of $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ solid solutions by explicitly enumerating the symmetry-independent configurations of various chalcogen compositions in small supercell models,⁵⁰ and we have more recently applied a similar approach to model the electronic structure of $\text{Ge}(\text{S}_{0.5}\text{Se}_{0.5})$.⁵¹ We have also developed an approach to predicting the κ_{latt} of solid solutions, which we successfully applied to $\text{Sn}(\text{S}_{0.1875}\text{Se}_{0.8125})$ and, subsequently, to explicit atomistic models of Ag-doped SnS.^{52,53} Another group have also developed a complementary approach using machine-learned interatomic potentials and molecular-dynamics simulations, and applied it to model the thermal transport in $\text{Sn}(\text{S}_{0.75}\text{Se}_{0.25})$.⁵⁴ However, the prediction of the S , σ and κ_{el} , which are required to evaluate the zT , has yet to be addressed.

In this work, we develop a complete workflow for predicting the zT of solid solutions and apply it to the *Pnma* $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ system, focusing on compositions close to the best-performing $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ identified in experiments. We compare our previous approach based on explicitly modelling the symmetry-independent configurations in a small supercell of the parent structure^{50,55} to using single disordered supercells with 32–256 atoms generated using the special quasi-random structure (SQS) method.⁵⁶ We combine our previous approach of modelling the lattice dynamics and κ_{latt} ⁵² with a methodology for calculating the electronic transport properties with approximate models for

the electron scattering rates.³⁹ We find that a single 128-atom SQS supercell provides an optimal trade-off between capturing the effects of disorder at the chalcogen site, as inferred from unfolded phonon dispersion curves, and providing realistic predictions of the electrical transport properties and zT at a manageable computational cost. Our calculations demonstrate that $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ can show lower κ_{latt} and comparable or larger maximum zT , zT_{max} , than the SnS and SnSe endpoints. In particular, we predict an averaged high-temperature n-type $zT_{\text{max}} = 2.4$, which is 50% larger than n-type SnS and comparable to n-type SnSe, and a room-temperature $zT_{\text{max}} = 0.62$ around 22% higher than SnSe and $2.4\times$ higher than SnS. Our methodology enables facile prediction of the thermoelectric performance of a wide range of homogeneous alloy thermoelectrics, and provides a route to using modelling to support experimental composition optimisation.

2 Computational methods

2.1 Alloy models

2.1.1 Systematic averaging over configurations

An initial model for the $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ alloy was constructed using the approach adopted in our previous studies.^{50,52,57}

The complete set of 23 symmetry-independent configurations in a 32-atom $2 \times 1 \times 2$ supercell of the *Pnma* SnS parent structure, with composition $\text{Sn}(\text{S}_{0.1875}\text{Se}_{0.8125})$, was generated using our in-house Transformer code.⁵⁸ Each of the $n = 23$ structures was fully optimised, and the lattice thermal conductivity, electrical-transport properties and zT were calculated as described below.

The occurrence probabilities P_i of each structure were then computed at a reference formation temperature $T_F = 900$ K according to:

$$P_i(T = T_F) = \frac{1}{Z(T_F)} g_i \exp[-E_i/(k_B T_F)] \quad (2)$$

E_i and g_i are the total energies and degeneracies of the structures, and k_B is the Boltzmann constant. $Z(T)$ is the thermodynamic partition function given by:

$$Z(T) = \sum_{i=1}^n g_i \exp[-E_i/(k_B T)] \quad (3)$$

We note that we are equating the E_i to the lattice energies U_i^{latt} , and that we do not account for the phonon contributions to the free energy, as we previously found these to have little impact on the distribution of P_i in this system with appropriately-converged phonon frequencies.⁵² Finally, weighted averages $\bar{X}$ of the calculated properties X were taken using the P_i according to:

$$\bar{X}(T = T_F) = \sum_{i=1}^n P_i(T_F) \times X_i \quad (4)$$

The main advantages of this approach are that it accounts for all the independent atomic configurations in the chosen supercell and provides access to thermodynamic information including the free energy of mixing and configurational entropy. On the other hand, the key downside is that the number of configurations

scales combinatorially with the supercell size, which severely restricts the size of the model that can be considered, and small supercells may not provide a realistic description of the medium- and long-range order in a real alloy.

2.1.2 The special quasi-random structure (SQS) method

As an alternative to the systematic averaging, we also explored the special quasi-random structure (SQS) method.⁵⁶

The idea behind the SQS approach is to represent a disordered material with a single, optimal supercell configuration. The occupations of the N alloying lattice sites i are denoted by a “spin” σ_i (after the Ising model, with $\sigma_i = \pm 1$ typically used for sites with two occupying atoms). The configuration is then discretised into geometric “figures” $f = (k, m)$ composed of k vertices separated by an m^{th} neighbour distance (e.g., $k = 2, m = 1$ corresponds to first-neighbour pairs). A series of correlation functions $\bar{\Pi}_{k,m}$ is then constructed from the average products of the σ_i over the lattice sites in the full set of symmetry-equivalent figures located at position $\mathbf{l}$. For a perfectly disordered binary system with composition x , the expectation values of the correlation functions are given by:

$$\langle \bar{\Pi}_{k,m} \rangle = (2x - 1)^k \quad (5)$$

The objective of the SQS is to arrange the alloying atoms in a chosen supercell such that the $\bar{\Pi}_{k,m}$ match the $\langle \bar{\Pi}_{k,m} \rangle$ for the perfectly random system as closely as possible.

The advantage of the SQS over systematic averaging is that it yields a single structure for a given composition and supercell size, but the downside is that a larger supercell is likely to be required in order to capture a sufficient diversity of local atomic environments. To this end, we generated SQS models with 32, 72, 128 and 256 atoms, corresponding to $2 \times 1 \times 2$, $3 \times 1 \times 3$, $4 \times 1 \times 4$ and $4 \times 2 \times 4$ expansions of the $Pnma$ SnS parent structure. The smaller 32- and 72-atom cells have chalcogen compositions of 81.3 and 80.6% Se, respectively, and the two larger cells both have 79.7% Se. Each of the four SQS structures was fully optimised and the thermoelectric properties were calculated in the same way as for the set of 32-atom supercell models generated with the systematic averaging approach, using the procedure outlined in the following subsection.

The SQS structures were generated using the Monte-Carlo algorithm⁵⁹ implemented in the Alloy Theoretic Automated Toolkit (ATAT),⁶⁰ taking into account pair clusters with a cutoff of 8.7 Å and triplet, quadruplet and quintuplet clusters with a cutoff of 6 Å to describe the disorder. Using this procedure, the correlation function of the 256-atom SQS model differed from that of the perfect random structure by < 0.1 , while those of the smaller 32- and 72/128-atom models differed by < 0.7 and < 0.2 respectively.

2.2 Prediction of zT

Our approach to predicting the zT of thermoelectric materials from first principles has been described extensively in our previous work^{42–44} and is briefly summarised here for completeness.

2.2.1 Phonon spectra and lattice thermal conductivity

The macroscopic κ_{latt} is the sum of the particle-like contribution κ_p and the wave-like contribution κ_w from intraband tunnelling:⁶¹

$$\kappa_{\text{latt}}(T) = \kappa_p(T) + \kappa_w(T) \quad (6)$$

The κ_p was calculated using the single mode-relaxation time approximation (SM-RTA), as a sum of contributions from phonon modes $\mathbf{q}j$, according to:³⁷

$$\kappa_p(T) = \frac{1}{NV} \sum_{\mathbf{q}j} \kappa_{\mathbf{q}j}(T) = \frac{1}{NV} \sum_{\mathbf{q}j} C_{\mathbf{q}j}(T) \mathbf{v}_{\mathbf{q}j} \otimes \mathbf{v}_{\mathbf{q}j} \tau_{\mathbf{q}j}(T) \quad (7)$$

where $\mathbf{q}$ and j are the phonon wavevector and band index, $C_{\mathbf{q}j}$, $\mathbf{v}_{\mathbf{q}j}$ and $\tau_{\mathbf{q}j}$ are the modal heat capacities, group velocities and lifetimes, V is the unit-cell volume, and N is the number of $\mathbf{q}$ included in the summation (equivalent to the number of unit cells in the crystal). The $C_{\mathbf{q}j}$ and $\mathbf{v}_{\mathbf{q}j}$ are calculated within the harmonic approximation using the 2nd-order interatomic force constants (FCs) $\Phi(2)$. The $\tau_{\mathbf{q}j}$ are the inverse of the phonon linewidths $\Gamma_{\mathbf{q}j}$:

$$\tau_{\mathbf{q}j}(T) = \frac{1}{\Gamma_{\mathbf{q}j}(T)} \quad (8)$$

The $\Gamma_{\mathbf{q}j}$ can be obtained perturbatively at third-order by combining the phonon frequencies $\omega_{\mathbf{q}j}$ and displacement vectors $\mathbf{W}_{\mathbf{q}j}$ calculated within the harmonic approximation with the 3rd-order (anharmonic) force constants $\Phi(3)$.³⁷

Computing the $\Phi(2)$ is relatively inexpensive, and it is therefore straightforward to calculate the $C_{\mathbf{q}j}$ and $\mathbf{v}_{\mathbf{q}j}$ of the various alloy models. However, the number of atomic displacements required to evaluate the $\Phi(3)$ increases rapidly with the cell size, and it is not practical to obtain the $\tau_{\mathbf{q}j}$ using this approach. For the majority of the alloy models we therefore instead adopt the method developed in our previous work.⁵² The phonon linewidths can be written approximately as:³⁷

$$\Gamma_{\mathbf{q}j}(T) \approx \frac{18\pi}{\hbar^2} P_{\mathbf{q}j} N_2(\mathbf{q}, \omega_{\mathbf{q}j}, T) \quad (9)$$

where $\hbar$ is the reduced Planck constant, $P_{\mathbf{q}j}$ is the averaged three-phonon interaction strength, and the function N_2 is a weighted two-phonon joint density of states (w-JDoS) that counts the number of energy- and momentum-conserving three-phonon scattering pathways available to a phonon with wavevector $\mathbf{q}$ and frequency $\omega_{\mathbf{q}j}$. The $P_{\mathbf{q}j}$ are given by:

$$P_{\mathbf{q}j} = \frac{1}{(3n_a)^2} \sum_{\mathbf{q}'j', \mathbf{q}''j''} |\Phi_{\mathbf{q}j, \mathbf{q}'j', \mathbf{q}''j''}|^2 \quad (10)$$

where n_a is the number of atoms in the primitive cell (i.e., there are $3n_a$ bands at each $\mathbf{q}$) and $\Phi_{\mathbf{q}j, \mathbf{q}'j', \mathbf{q}''j''}$ are the three-phonon interaction strengths.³⁷ The N_2 are calculated by summing the number of pathways for collision (Type 1) and decay (Type 2) processes:³⁷

$$N_2(\mathbf{q}, \omega, T) = N_2^{(1)}(\mathbf{q}, \omega, T) + N_2^{(2)}(\mathbf{q}, \omega, T) \quad (11)$$

$$N_2^{(1)}(\mathbf{q}, \omega, T) = \frac{1}{N} \sum_{\mathbf{q}', \mathbf{q}'', \mathbf{q}'''} \Delta(\mathbf{q}, \mathbf{q}', \mathbf{q}'') [n_{\mathbf{q}'j'}(T) - n_{\mathbf{q}''j''}(T)] \times [\delta(\omega + \omega_{\mathbf{q}'j'} - \omega_{\mathbf{q}''j''}) - \delta(\omega - \omega_{\mathbf{q}'j'} + \omega_{\mathbf{q}''j''})] \quad (12)$$

$$N_2^{(2)}(\mathbf{q}, \omega, T) = \frac{1}{N} \sum_{\mathbf{q}', \mathbf{q}'', \mathbf{q}'''} \Delta(\mathbf{q}, \mathbf{q}', \mathbf{q}'') [n_{\mathbf{q}'j'}(T) + n_{\mathbf{q}''j''}(T) + 1] \times \delta(\omega - \omega_{\mathbf{q}'j'} - \omega_{\mathbf{q}''j''}) \quad (13)$$

The functions δ and Δ enforce the conservation of energy and (crystal) momentum, respectively, for three-phonon scattering processes, and the $n_{\mathbf{q}j}$ are the phonon occupation numbers from the Bose-Einstein distribution:

$$n_{\mathbf{q}j}(T) = \frac{1}{\exp[\hbar\omega_{\mathbf{q}j}/(k_B T)] - 1} \quad (14)$$

The N_2 depend only on the $\omega_{\mathbf{q}j}$ and are therefore calculable within the harmonic approximation. The $\Gamma_{\mathbf{q}j}$ can therefore be obtained without explicit calculation of the $\Phi(3)$ by replacing the $P_{\mathbf{q}j}$ in Eq. (9) with a constant value $\bar{P}$. Suitable values for the alloys were obtained by linear interpolation and appropriate scaling of the $\bar{P}$ determined for the *Pnma* SnS and SnSe endpoints in previous work.⁵²

While SM-RTA model in Eq. 7 provides a closed-form solution to the phonon Boltzmann transport equation (BTE) for the κ_p , obtaining the κ_w requires solving the Wigner transport equation (WTE).⁶¹ We have previously shown that the κ_w term makes an important contribution to the κ_{latt} of the structurally-complex “ π ” phases of SnS and SnSe and “dilute” supercell models of Ag-doped SnS.^{44,53} In particular, we found that accounting for this contribution is necessary to recover the convergence of the κ_{latt} of Ag-doped SnS to the bulk value as the doping level is reduced by progressively increasing the supercell size.⁵³ Therefore, while we did not test this in our previous work on the Sn(S_{0.2}Se_{0.8}) alloy,⁵² we would expect the wave-like conductivity to be important for the alloys, and we calculated the κ_w by solving the WTE⁶¹ using the same $\tau_{\mathbf{q}j}$ as for the κ_p .

2.2.2 Electronic transport properties

The three electrical transport properties $\mathcal{S}$, σ and κ_{el} can be obtained in a similar manner to the particle-like contribution to the κ_{latt} in Eq. (7), by solving the electron BTE within the momentum relaxation-time approximation (MRTA).^{38,39}

We first calculate the spectral conductivity Σ and generalised moments of the transport distribution function $L^n(T)$ given by:

$$\Sigma(\varepsilon, T) = \sum_j \int \frac{1}{8\pi^3} \mathbf{v}_{\mathbf{k}j} \otimes \mathbf{v}_{\mathbf{k}j} \tau_{\mathbf{k}j}(T) \delta[\varepsilon - \varepsilon_{\mathbf{k}j}] d\mathbf{k} \quad (15)$$

$$L^n(\varepsilon_F, T) = e^2 \int \Sigma(\varepsilon, T) (\varepsilon - \varepsilon_F)^n \left[-\frac{\partial f^0(\varepsilon, \varepsilon_F, T)}{\partial \varepsilon} \right] d\varepsilon \quad (16)$$

where e is the elementary charge, $\mathbf{k}$ and j are the electron wavevectors and band indices, respectively, and $\mathbf{v}_{\mathbf{k}j}$, $\tau_{\mathbf{k}j}$ and $\varepsilon_{\mathbf{k}j}$ are the electron group velocities, lifetimes and band energies.

$f^0(\varepsilon, T)$ is the Fermi-Dirac distribution function given by:

$$f^0(\varepsilon, \varepsilon_F, T) = \frac{1}{\exp[(\varepsilon - \varepsilon_F)/k_B T] + 1} \quad (17)$$

where ε_F is the Fermi energy. The $\varepsilon_{\mathbf{k}j}$ and $\mathbf{v}_{\mathbf{k}j}$ can be obtained from a high-quality electronic structure calculation. The $\tau_{\mathbf{k}j}$ can be obtained in a similar manner to the $\tau_{\mathbf{q}j}$ for phonons, i.e., by considering three-particle interactions between electrons and phonons, from electron-phonon coupling calculations. However, these are currently difficult to access and not widely used, particularly for structurally-complex systems such as alloys. Instead, we adopt the approach in Ref. 39, where the $\tau_{\mathbf{k}j}$ are calculated from phenomenological expressions for common “universal” semiconductor scattering mechanisms, viz. acoustic deformation potential (ADP), polar-optic phonon (POP), ionised impurity (IMP) and piezoelectric (PIE) scattering:

$$\tau_{\mathbf{k}j}^{-1} = \left(\tau_{\mathbf{k}j}^{\text{ADP}} \right)^{-1} + \left(\tau_{\mathbf{k}j}^{\text{POP}} \right)^{-1} + \left(\tau_{\mathbf{k}j}^{\text{IMP}} \right)^{-1} + \left(\tau_{\mathbf{k}j}^{\text{PIE}} \right)^{-1} \quad (18)$$

The σ , $\mathcal{S}$ and κ_{el} are then calculated from the L^n as:

$$\sigma(\varepsilon_F, T) = L^0(\varepsilon_F, T) \quad (19)$$

$$\mathcal{S}(\varepsilon_F, T) = \frac{1}{eT} \frac{L^1(\varepsilon_F, T)}{L^0(\varepsilon_F, T)} \quad (20)$$

$$\kappa_{\text{el}}(\varepsilon_F, T) = \frac{1}{e^2 T} \left[\frac{(L^1(\varepsilon_F, T))^2}{L^0(\varepsilon_F, T)} - L^2(\varepsilon_F, T) \right] \quad (21)$$

Under the assumption that chemical doping has a minimal effect on the electronic band structure, the ε_F can be adjusted to impose an extrinsic carrier concentration (“doping level”) n (termed the “rigid band” approximation).^{62,63} This approach therefore allows the electrical properties to be predicted as a function of n and T , and we have previously obtained accurate results for *Pnma* SnS and SnSe and for other related Group IV-VI monochalcogenides.^{42–44}

In practice, this approach to predicting the electrical properties requires a high-quality electronic-structure calculation to obtain an accurate electronic bandgap E_g and band energies $\varepsilon_{\mathbf{k}j}$, together with calculations of the various material properties required to estimate the $\tau_{\mathbf{k}j}$.³⁹ One would typically use a hybrid functional (e.g., HSE06^{64,65}) to compute the E_g and $\varepsilon_{\mathbf{k}j}$, but this can be challenging for large unit cells and high $\mathbf{k}$ -point densities. To mitigate this, we adopt the non-self-consistent hybrid approach developed in our previous work,⁶⁶ where we compute both quantities using the Kohn-Sham orbitals from a PBE⁶⁷ calculation.

2.2.3 Thermoelectric figure of merit

Finally, the $\kappa_{\text{latt}}(T)$ can be combined with the $\mathcal{S}(n, T)$, $\sigma(n, T)$ and $\kappa_{\text{el}}(n, T)$ to estimate the figure of merit zT as a function of doping level and temperature. The four properties entering the Eq. (1) are tensor quantities, and the figure of merit is therefore orientation dependent. The zT along a particular Cartesian direction $\alpha = x, y, z$ can be calculated as:

$$(zT)^{\alpha\alpha} = \frac{(S^{\alpha\alpha})^2 \sigma^{\alpha\alpha}}{\kappa_{\text{latt}}^{\alpha\alpha} + \kappa_{\text{el}}^{\alpha\alpha}} \quad (22)$$

This mirrors the way in which the figure of merit is calculated from experimental measurements of the constituent properties. Alternatively, for pressed pellets with randomly-oriented crystallites, the four properties can be replaced by their scalar averages S , σ , κ_{latt} and κ_{el} given by e.g.,:

$$S = \frac{1}{3} \text{Tr}[\mathbf{S}] = \frac{1}{3} (S^{\text{xx}} + S^{\text{yy}} + S^{\text{zz}}) \quad (23)$$

We note that the power factors $S^2\sigma$ in the numerator of the zT equation are calculated in an analogous manner, i.e., from the $S^{\alpha\alpha}$ and $\sigma^{\alpha\alpha}$ along a given direction or from the scalar-averaged S and σ .

2.3 First-principles calculations

DFT calculations were carried out using periodic pseudopotential plane-wave density-functional theory (DFT) as implemented in the Vienna *Ab initio* Simulation Package (VASP) code.⁶⁹

Electron exchange and correlation were treated using the dispersion-corrected PBEsol+D3 functional^{70,71} used in several of our previous studies on these and similar materials.^{43,44,50,72,73} The core electrons were modelled using projector-augmented wave (PAW) pseudopotentials^{74,75} with the Sn 5s/5p, the S 3s/3p, and Se 4s/4p electrons in the valence region. We note that we have previously used a Sn pseudopotential including the semi-core Sn 4d electrons in the valence region, but we found that incorporating these into the pseudopotential core significantly reduced the computational cost of the calculations while having a minimal effect on the total energy, optimised lattice parameters, and electronic bandgap (Table S1[†]). The valence electronic structure was modelled using a plane-wave basis with a cutoff of 600 eV and the Γ -centred Monkhorst-Pack $\mathbf{k}$ -point meshes⁶⁸ listed in Table 1.

All structures were fully optimised to strict tolerances of 10^{-8} eV on the total energy and 10^{-2} eV $\text{\AA}^{-1}$ on the forces. Second-order (harmonic) force constants $\Phi(2)$ and, for selected systems, third-order (anharmonic) force constants $\Phi(3)$ were computed using the supercell finite-displacement method using the Phonopy³⁶ and Phono3py³⁷ codes, respectively. The $\Phi(2)$ and $\Phi(3)$ were calculated using the default displacement steps of 10^{-2} and 3×10^{-2} $\text{\AA}$ and with the supercells, reduced $\mathbf{k}$ -point sampling, and, where appropriate, interatomic distance cutoffs listed in Table 1. Phonon density of states curves $g(f)$ were obtained by interpolating the frequencies onto the uniform Γ -centred $\mathbf{q}$ -point meshes listed in Table 1. Phonon dispersions were obtained by using the band-unfolding approach in Ref. 76 to project phonon modes back to the Brillouin zone of the closest *Pnma* SnSe endpoint. Non-analytical corrections were applied to describe splitting between longitudinal and transverse optic modes (LO/TO splitting) using the method in Ref. 77 and with high-frequency dielectric constants ϵ_{∞} and Born charges $\mathbf{Z}^*$ computed using density-functional perturbation theory (DFPT).⁷⁸ For the smaller 32- and 72-atom SQS models, the phonon lifetimes were computed by combining the harmonic frequencies and eigenvectors from with the $\Phi(3)$ calculated as outlined in Table 1. For these and the larger models the lifetimes were (also) estimated

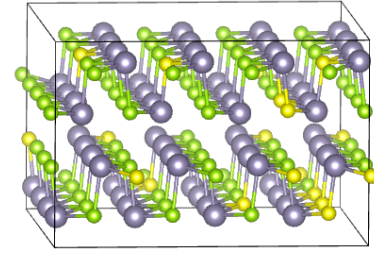


Fig. 1 Optimised structure of the 128-atom special quasi-random structure (SQS) model of $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$. The Sn, S and Se atoms are shown in purple, yellow and green respectively. This image was generated using VESTA software.⁸²

using the approach in our previous work⁵² with the scaled average three-phonon interaction strengths in Table 1. The κ_{latt} was evaluated using modal properties computed on the $\mathbf{q}$ -point grids listed in Table 1.

The electronic transport properties were obtained using the AMSET code.³⁹ Accurate band gaps E_g and high-frequency dielectric constants ϵ_{∞} were obtained using non-self-consistent HSE06^{65,66} in conjunction with the linear-optics routines in VASP.⁷⁸ These E_g were then used to correct uniform band structures evaluated on the “dense” $\mathbf{k}$ -point meshes listed in Table 1. We note that while the typical $2\times$ enhancement factor used in previous work was found to be sufficient for the smaller 32- and 72-atom models, a larger $3\times$ enhancement was required for the larger 128- and 256-atom models to accurately capture the dispersion and carrier effective masses (Table S2 and Figs. S1-S4[†]). AMSET calculates electronic relaxation times by considering four scattering mechanisms, *viz.* acoustic deformation potential (ADP), piezoelectric (PIE), polar optical phonon (POP) and ionised impurity (IMP) scattering, which require the calculation of several other properties. The band deformation potentials $D_{\mathbf{k}j}$ were computed by performing a series of single-point energy calculations on deformed structures, generated with AMSET, using the “base” $\mathbf{k}$ -point meshes. The elastic constant matrices $\mathbf{C}$ were obtained using the finite differences approach in VASP.⁷⁹ The static dielectric constants $\epsilon_S = \epsilon_{\infty} + \epsilon_{\text{ionic}}$ were obtained by combining the HSE06 ϵ_{∞} with the ϵ_{ionic} obtained using the calculated phonon frequencies and eigenvectors at $\mathbf{q} = \Gamma$, the $\mathbf{Z}^*$, and an implementation of the procedure in Ref. 80 in our Phonopy-Spectroscopy package.⁸¹ Finally, the piezoelectric stress tensor $\mathbf{e}$ was computed from the same DFPT approach as used to obtain the $\mathbf{Z}^*$ for the non-analytical correction to the phonon frequencies.⁷⁸

3 Results and discussion

3.1 Structure, dynamics and lattice thermal conductivity

The optimised structure of the 128-atom SQS model of *Pnma* $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ is shown in Fig. 1, and those of the 32-, 72- and 256-atom SQS models are shown in Fig. S5.[†] The corresponding optimised lattice constants, referenced back to the parent *Pnma* structure, are shown in Table 2. For the four SQS models, these are simply the optimised lattice parameters divided by the supercell expansion along the corresponding direction. For the 32-atom

Table 1 Technical parameters used in the calculations performed in this work. For each of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models we examine, we list the “base” and “dense” k -point sampling meshes used for selected electronic-structure calculations (see text), the supercell and reduced k -point sampling used to evaluate the second-order force constants $\Phi(2)$, the supercell, reduced k -point sampling, range cutoff and estimated average three-phonon interaction strengths $\bar{P}$ used to evaluate the third-order force constants $\Phi(3)$, where applicable, and the q -point sampling used to compute the phonon density of states $g(f)$ and the lattice thermal conductivity κ_{latt} . k - and q -point meshes are specified as the number of subdivisions on a regular Γ -centred Monkhorst-Pack grid.⁶⁸

	k -Point Sampling		$\Phi(2)$	$\Phi(3)$			q -Point Sampling	
	Base	Dense	Supercell (# Atoms)	k -Points	Cutoff / Å	$\bar{P}$ / eV ²	$g(\omega)$	κ_{latt}
32-atom (enum)	$4 \times 3 \times 5$	$8 \times 6 \times 10$	$2 \times 2 \times 2$ (256)	$2 \times 2 \times 3$	-	1.47×10^{-12}	$24 \times 24 \times 24$	$9 \times 9 \times 9$
32-atom (SQS)	$4 \times 3 \times 5$	$8 \times 6 \times 10$	$2 \times 2 \times 2$ (256)	$2 \times 2 \times 3$	-	1.47×10^{-12}	$24 \times 24 \times 24$	$9 \times 9 \times 9$
72-atom (SQS)	$3 \times 3 \times 3$	$6 \times 6 \times 6$	$2 \times 2 \times 2$ (576)	$2 \times 2 \times 2$	4.5	2.93×10^{-13}	$24 \times 24 \times 24$	$11 \times 11 \times 11$
128-atom (SQS)	$2 \times 3 \times 3$	$6 \times 9 \times 9$	$1 \times 2 \times 1$ (256)	$2 \times 2 \times 3$	-	9.36×10^{-14}	$12 \times 24 \times 12$	$11 \times 11 \times 11$
256-atom (SQS)	$2 \times 2 \times 3$	$6 \times 6 \times 9$	$1 \times 1 \times 1$ (256)	$2 \times 2 \times 3$	-	2.34×10^{-14}	$12 \times 12 \times 12$	$11 \times 11 \times 11$

Table 2 Optimised lattice parameters and calculated formation enthalpies ΔH_F of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work. The lattice parameters are provided in terms of the parent $Pnma$ SnS/SnSe structures and compared to values for the endpoints from similar calculations.⁴² Unless otherwise stated the properties for the 32-atom supercell enumeration model are calculated for a formation temperature $T_F = 900$ K.

	a [Å]	b [Å]	c [Å]	V [Å ³]	ΔH_F [meV per F.U.]
SnS	4.22	10.99	3.96	183.66	-
$\text{Sn}(\text{S}_{0.187}\text{Se}_{0.813})$ (32-atom, enum)	4.30	11.30	4.09	198.96	5.87 (5.39 at $T_F = 10$ K)
$\text{Sn}(\text{S}_{0.187}\text{Se}_{0.813})$ (32-atom, SQS)	4.30	11.30	4.10	199.22	5.90
$\text{Sn}(\text{S}_{0.194}\text{Se}_{0.806})$ (72-atom, SQS)	4.30	11.29	4.09	198.56	5.86
$\text{Sn}(\text{S}_{0.203}\text{Se}_{0.797})$ (128-atom, SQS)	4.30	11.28	4.09	198.38	5.79
$\text{Sn}(\text{S}_{0.203}\text{Se}_{0.797})$ (256-atom, SQS)	4.30	11.28	4.09	198.38	4.43
SnSe	4.34	11.35	4.12	202.95	-

supercell enumeration model the optimised lattice parameters are first averaged according to Eq. (2) for a typical alloy formation temperature $T_F = 900$ K, and then divided by the supercell expansion. The lattice parameters obtained for all five models are very similar and fall between those of the SnS and SnSe endpoints, and closer to SnSe, as expected from Vegard’s law.

Table 2 also shows the calculated formation enthalpies ΔH_F calculated according to:

$$\Delta H_F = E_{\text{Sn}(\text{S}_{1-x}\text{Se}_x)} - [(1-x)E_{\text{SnS}} + xE_{\text{SnSe}}] \quad (24)$$

where E are the DFT total energies of the alloy and SnS/SnSe endpoints, which are equivalent to the enthalpies H when relaxed to $p \approx 0$. For the 32-atom supercell enumeration model, the E for the alloy is again obtained from a thermodynamic average, and we obtain values of $\Delta H_F = 5.39$ and 5.87 meV per F.U. respectively at $T_F = 10$ and 900 K. The latter is very similar to the values of $\Delta H_F = 5.79$ – 5.9 meV per F.U. obtained for the 32-, 72- and 128-atom SQS models, whereas the low-temperature ΔH_F lies between those of the three smaller SQS cells and the lower $\Delta H_F = 4.43$ meV per F.U. obtained for the larger 256-atom model. Nevertheless, all five models yield the same qualitative prediction of a small positive formation enthalpy, which is in line with previous calculations.^{50,83}

At the typical (higher) $T_F = 900$ K we predict a formation free energy $\Delta G_F = -24.8$ meV per F.U. and a configurational entropy of $TS_{\text{conf}} = 30.7$ meV per F.U. for the 32-atom supercell enumeration model. Under the assumption of ideal mixing behaviour, a “first-order” value of TS_{conf} for the four SQS models can be estimated from:

$$TS_{\text{conf}} = -RT[x \ln x + (1-x) \ln(1-x)] \quad (25)$$

These ideal values range from 37.4–39.1 meV per F.U., which are

24–28% larger than that calculated for the supercell enumeration model. The ΔH_F are therefore sufficiently large in this system that an explicit average over configurations is necessary for a correct description of the thermodynamics. Nonetheless, we note that the predicted ΔH_F is well below $k_B T \approx 25$ meV atom⁻¹ at room temperature, and so we would not expect spontaneous phase separation except at very low temperatures where the atomic diffusion would likely be too slow for this to be kinetically feasible.

The unfolded phonon dispersion and atom-projected density of states (PDoS) curves of the five alloy models are compared in Fig. 2. There are no imaginary modes in the unfolded phonon dispersions of any of the alloy models, which confirms that all 23 structures in the 32-atom supercell enumeration model, and all four of the SQS models, are dynamically stable.

As found in previous work,⁵² a characteristic effect of alloying at the chalcogen site is a “smearing” of the high-frequency phonon bands associated with motion of the chalcogen sublattice. This is clearly seen in the (averaged) unfolded dispersion for the 32-atom supercell enumeration model, whereas the 32-atom SQS model clearly shows more pronounced fine structure in this part of the spectrum. The 72-atom SQS model partially captures the smearing but retains some discernible fine structure, while the larger 128- and 256-atom SQS show strong smearing of the high-frequency bands similar to the supercell-enumeration model. Similar contrasts can also be seen through the presence or absence of sharp features in the (P)DoS curves of the five models. Most notably, while the larger SQS cells appear to effectively converge the shape of the low- and mid-frequency features in the DoS, attributable to Sn and Se respectively, they retain some fine structure in the high-frequency feature from the S-based modes that is averaged out in the supercell enumeration model. We attribute this to the fact that there are comparatively few S atoms in

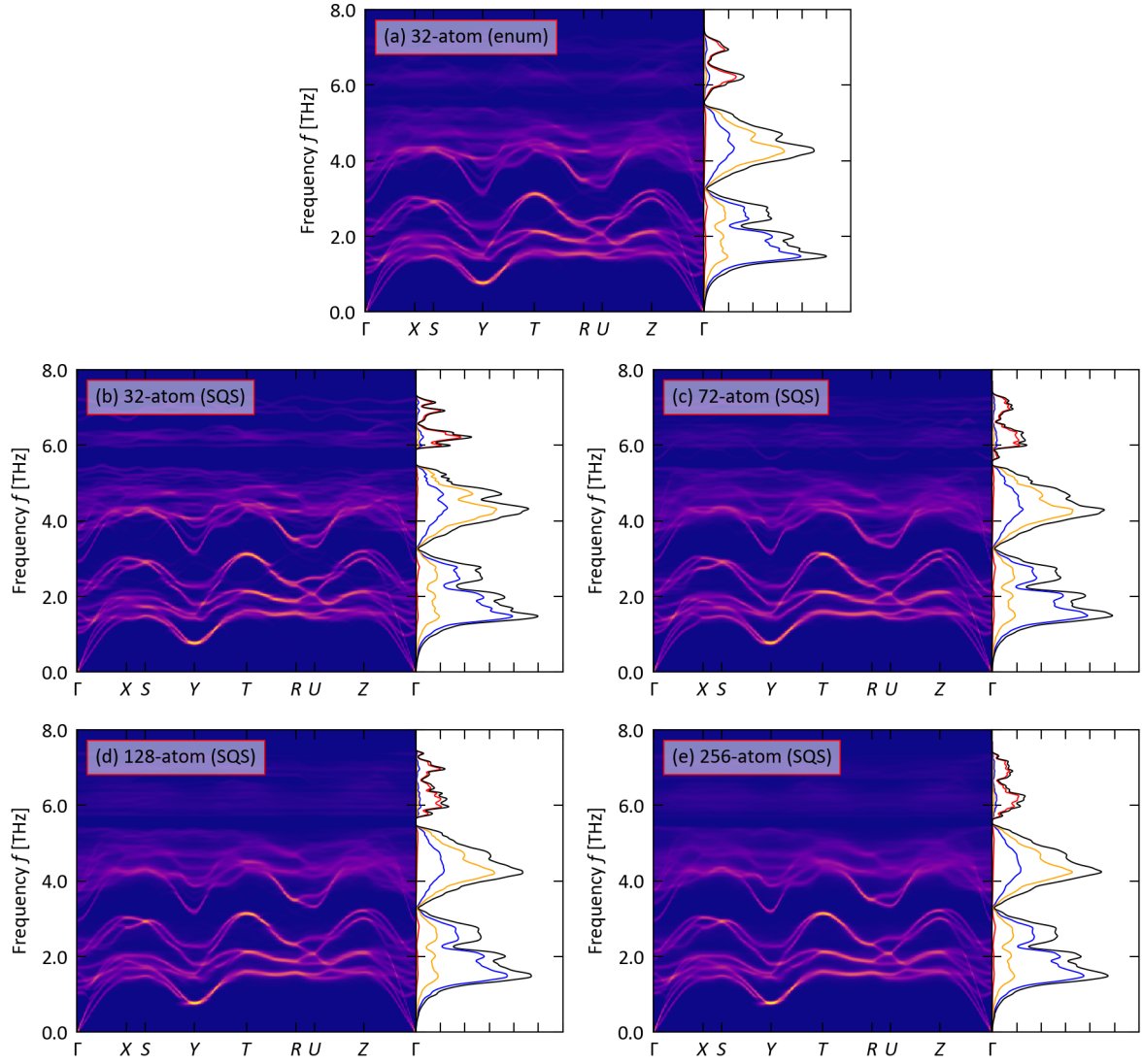


Fig. 2 Unfolded dispersion and density of states (DoS) curves of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work. The dispersion and PDoS of the 32-atom supercell enumeration model are calculated for a formation temperature $T_F = 900$ K. The colour scale on the phonon dispersion plots shows the spectral weight and ranges from blue (small weight) to yellow (large weight). On each DoS plot the total DoS is shown in black and the eigenvector projections onto the S, Se and Sn atoms are shown in red, orange and blue respectively.

our chosen composition with $x \approx 0.8$, which means a single small supercell model cannot capture the range of S environments in the alloy.

The lattice thermal conductivity κ_{latt} arises from two fundamental mechanisms, viz. “particle-like” transport κ_p and a “wave-like” transport κ_w .⁶¹ The κ_p is sensitive to the phonon group velocities $\mathbf{v}_q$ and is strongly suppressed by the symmetry breaking induced by alloying. On the other hand, the κ_w depends inversely on the interband frequency spacing and is enhanced in the alloys due to the band smearing. We have previously determined that these two mechanisms can partially compensate for one another, and both must in principle be considered to obtain an accurate prediction of the κ_{latt} .⁵³

The “pseudo 2D” structure of orthorhombic SnS and SnSe leads to strong anisotropy in the κ_{latt} , with larger in-plane conductivity

along the crystallographic a and c directions and much smaller conductivity along the layering b direction. However, for typical polycrystalline pellets and thin films, the scalar average κ_{latt} is more representative:

$$\kappa_{\text{latt}} = \frac{1}{3} \text{Tr}[\boldsymbol{\kappa}_{\text{latt}}] = \frac{1}{3} [\kappa_{\text{latt}}^{xx} + \kappa_{\text{latt}}^{yy} + \kappa_{\text{latt}}^{zz}] \quad (26)$$

where Tr denotes the matrix trace (the sum of the diagonal elements). Equivalent averaging can be applied to both the κ_p and κ_w .

The predicted κ_{latt} for the five alloy models examined in this work are compared to previous calculations on SnS and SnSe⁵² in Fig. 3. There is a substantial reduction in the κ_{latt} of all five alloys with respect to both endpoints, with a particularly pronounced effect at lower temperature. We obtain room-temperature ($T = 300$ K) $\kappa_{\text{latt}} = 0.75\text{--}0.94 \text{ W m}^{-1} \text{ K}^{-1}$ for the alloy models, which

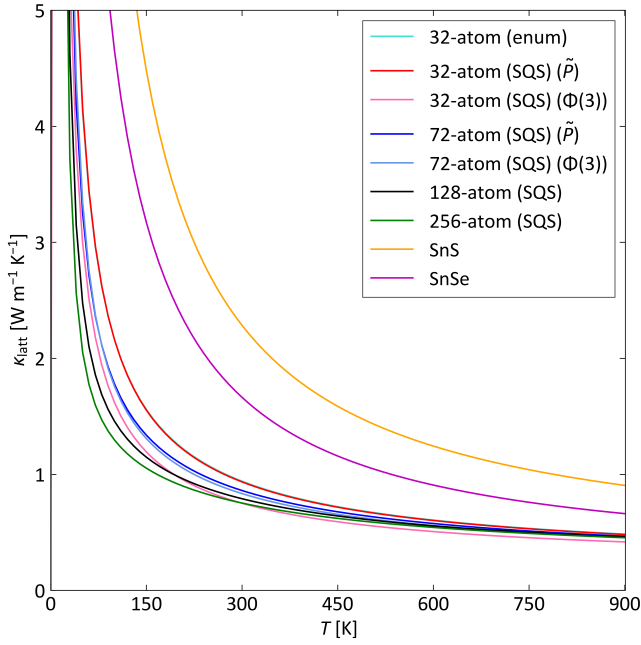


Fig. 3 Scalar-average total lattice thermal conductivity $\kappa_{\text{latt}} = \kappa_p + \kappa_w$, including both the “particle-like” and “wave-like” contributions κ_p/κ_w , of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work. Data for SnS and SnSe from our previous studies^{42,44,52} are also shown for comparison.

are significantly smaller than the $\kappa_{\text{latt}} = 2.28$ and $1.66 \text{ W m}^{-1} \text{ K}^{-1}$ calculated for SnS and SnSe. These values are reduced to $\kappa_{\text{latt}} = 0.47\text{--}0.56 \text{ W m}^{-1} \text{ K}^{-1}$ at $T = 700 \text{ K}$, which still represent a favourable reduction compared to the endpoints ($1.10/0.80 \text{ W m}^{-1} \text{ K}^{-1}$). We note that the room-temperature κ_{latt} for the 32-atom supercell enumeration model is larger than that obtained in our previous work,⁵² which is primarily because we did not include the κ_w .

Our prediction of reduced κ_{latt} in the alloys aligns well with experimental studies on these systems.^{33–35} All three studies are on consolidated powders, for which the consolidation process typically results in a strong preferred crystallite orientation with the layering and in-plane directions respectively parallel and perpendicular to the pressing direction. In this case, the scalar average in Eq. (26) is no longer appropriate, and the more realistic comparisons with the calculations are:

$$\kappa_{\text{latt}}^{\parallel} = \kappa_{\text{latt}}^{\text{yy}} \quad (27)$$

$$\kappa_{\text{latt}}^{\perp} = \frac{1}{2} [\kappa_{\text{latt}}^{\text{xx}} + \kappa_{\text{latt}}^{\text{zz}}] \quad (28)$$

Ref. 34 measured a value of $\kappa_{\text{latt}}^{\parallel} = 0.52 \text{ W m}^{-1} \text{ K}^{-1}$ for $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ at 323 K , which is within the range of our predictions of $0.46\text{--}0.56 \text{ W m}^{-1} \text{ K}^{-1}$ at $T = 320 \text{ K}$. On the other hand, Ref. 33 quotes a $\kappa_{\text{latt}}^{\parallel}$ of $0.11 \text{ W m}^{-1} \text{ K}^{-1}$ for $(\text{Ag}_{0.01}\text{Sn}_{0.99})(\text{S}_{0.15}\text{Se}_{0.85})$ at $T = 825 \text{ K}$, which is substantially lower than our calculated $0.30\text{--}0.34 \text{ W m}^{-1} \text{ K}^{-1}$ at a comparable $T = 830 \text{ K}$. This discrepancy could be attributed to a number of factors, including a finite grain size and/or the presence of defects in the experimental measurements, or to our model failing to cap-

ture the high-temperature anharmonicity in these materials.^{13,28} Nonetheless, the κ_{latt} calculated for the alloys are lower than the $\kappa_{\text{latt}} = 0.47$ we predict for bulk SnSe, and our calculations therefore do capture the qualitative effect of alloying. A comparison of the $\kappa_{\text{latt}}^{\parallel}$ and $\kappa_{\text{latt}}^{\perp}$ to the scalar-averaged κ_{latt} computed using Eq. (26) is provided in Fig. S6.[†]

The κ_{latt} of the 32-atom supercell enumeration model and the larger 72-, 128- and 256-atom models were computed using the approach in Ref. 52, where the three-phonon interaction strengths $\Phi_{\mathbf{q}\mathbf{j},\mathbf{q}'\mathbf{j}',\mathbf{q}''\mathbf{j}''}$ are set to a constant value $\tilde{P}$ estimated from the endpoint structures.⁵² The $\tilde{P}$ are determined from a linear fit of the κ_{latt} at a representative temperature, and are typically temperature-independent or weakly T -dependent above $\sim 300 \text{ K}$.^{52,53,84} This procedure allows the predicted κ_{latt} to take into account differences in the harmonic modal properties and spectra, the latter of which affects the phonon lifetimes $\tau_{\mathbf{q}\mathbf{j}}$ through conservation of energy selection rules, but avoids the intractable cost of computing the third-order force constants $\Phi(3)$ for large supercells.

For the smaller 32- and 72-atom SQS models it was also feasible to compute the $\Phi(3)$, albeit using a range cutoff of $4.5 \text{ \AA}$ for the larger model, and to calculate the Φ explicitly, allowing us to directly validate this approach. The $\tilde{P}$ estimated for these models by linear interpolation between the endpoints are $1.47 \times 10^{-12} \text{ eV}^2$ and $2.93 \times 10^{-13} \text{ eV}^2$, respectively, which are reasonably close to the values of $2.11 \times 10^{-12} \text{ eV}^2$ and $3.10 \times 10^{-13} \text{ eV}^2$ determined by fitting the κ_{latt} obtained from the explicit calculations. The values of the κ_{latt} at 300 K (i.e., the fitting temperature) obtained for the two models using the two sets of $\tilde{P}$ are $0.75/0.94$ and $0.83/0.86 \text{ W m}^{-1} \text{ K}^{-1}$ respectively, and the estimated values are both within 10% of the explicit calculations. Given the typical variability in experimental measurements, we consider this level of accuracy more than acceptable, and we therefore consider our method to be a viable approach to estimating the κ_{latt} of alloy models in cases where it is not feasible to compute the $\Phi(3)$.

Finally, to understand the impact of alloying on the κ_{latt} of the alloys, we compare the contributions from the κ_p and κ_w (Fig. 4(a)/(b)). The κ_p systematically decreases as the model size increases from 32 to 256 atoms. On the other hand, the κ_w shows the opposite behaviour, with the result that the overall κ_{latt} of all five alloy models are similar. The behaviour of the κ_p can be understood using the analysis developed in our previous work.⁹ We write the κ_p as the product of a harmonic function $\kappa_p/\tau_{\text{ph}}$ and a weighted-average lifetime τ_{ph} according to:

$$\kappa = \tau_{\text{ph}} \times \frac{1}{N_q V} \sum_{\mathbf{q}\mathbf{j}} \frac{\kappa_{\mathbf{q}\mathbf{j}}}{\tau_{\mathbf{q}\mathbf{j}}} = \tau_{\text{ph}} \times \frac{1}{N_q V} \sum_{\mathbf{q}\mathbf{j}} C_{\mathbf{q}\mathbf{j}} \mathbf{v}_{\mathbf{q}\mathbf{j}} \otimes \mathbf{v}_{\mathbf{q}\mathbf{j}} \quad (29)$$

The $\kappa_p/\tau_{\text{ph}}$ predominantly reflect differences in the phonon group velocities, and show a systematic decrease with model size, which mirrors the behaviour of the κ_p (Fig. 4(c)). On the other hand, the majority of the τ_{ph} span a narrow range, with those of the 32-atom SQS model with explicit $\Phi(3)$ and the 256-atom SQS model with estimated $\tilde{P}$ being notable outliers (Fig. 4(d)). This indicates that the reduction in κ_p is primarily due to lower phonon group velocities. For the pure chalcogenides this quantity shows

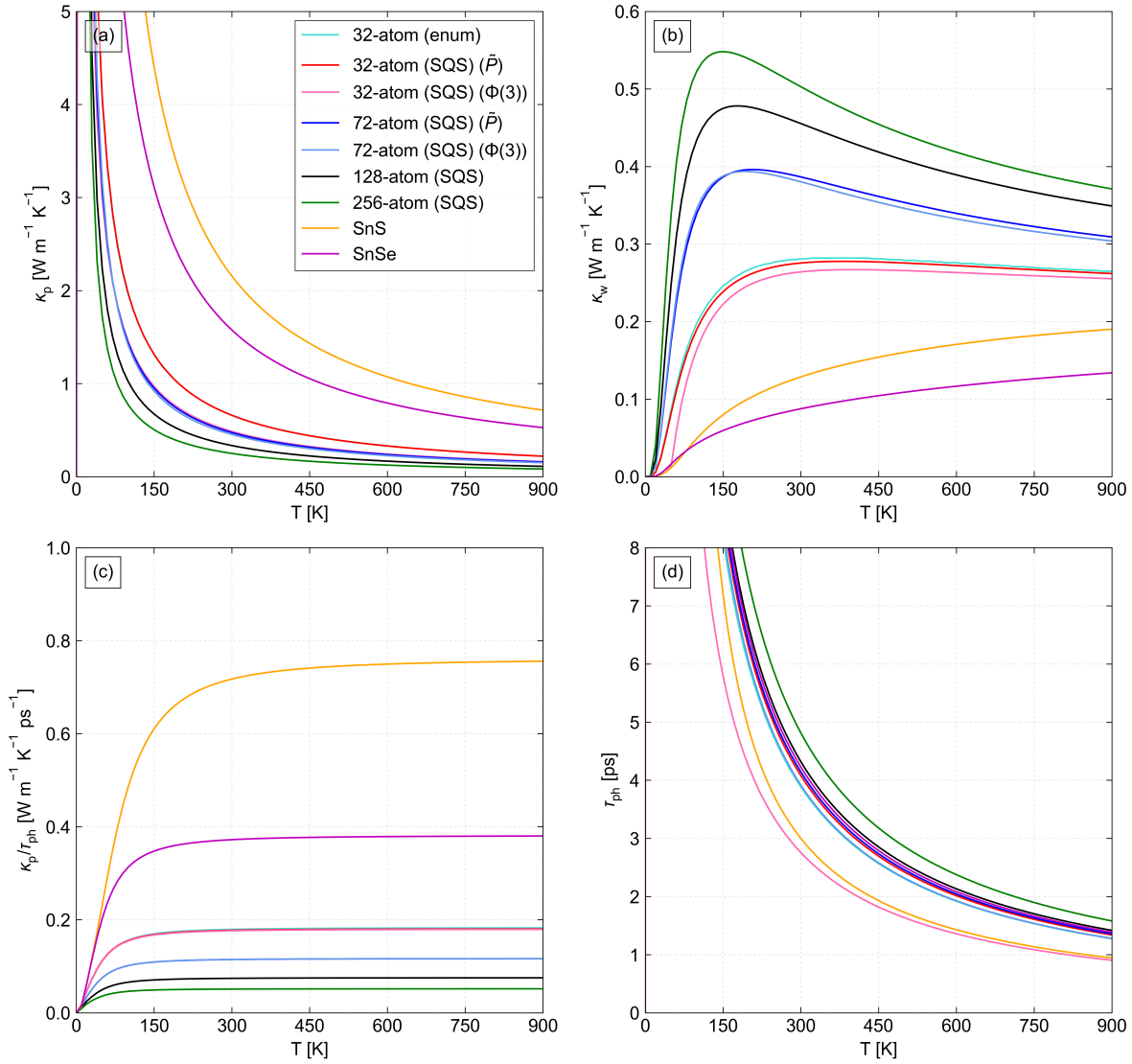


Fig. 4 Analysis of the scalar-average thermal conductivity κ_{latt} of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work. Plots (a) and (b) show the contributions to the κ_{latt} from “particle-like” transport κ_p (a) and “wave-like” transport κ_w (b). Plots (c) and (d) show an analysis of the κ_p in terms of a harmonic term $\kappa_p/\tau_{\text{ph}}$ (c) and a weighted-average phonon lifetime τ_{ph} (d) as described in Eq. 8 in the text.⁹ For the 32- and 72-atom special quasi-random structure (SQS) models we show analyses of calculations with explicit third-order force constants $\Phi(3)$ and using estimated average three-phonon interaction strengths $\tilde{P}$. Data for SnS and SnSe from our previous studies^{42,44,52} are shown for comparison.

a strong inverse correlation with unit-cell size,⁹ attributed to the less regular bonding environment in more structurally-complex materials, so this result is not surprising. On the other hand, the reduction in artificial/imposed periodicity results in a smaller Brillouin zone and substantial band folding, narrowing the inter-band spacing and increasing the κ_w . Our calculations therefore suggest that increasing the level of disorder in alloys lowers the phonon velocities and the particle-like thermal conductivity, but that this is compensated by an increase in wave-like transport. This mirrors the finding in our previous work on Ag-doped SnS,⁵³ where the same behaviour reproduced the expected convergence of the κ_{latt} towards the bulk limit when the dopant was diluted by using progressively larger simulation cells, and underscores the importance of accounting for this contribution in calculations on

alloys.

3.2 Electronic transport properties

The calculated band gaps E_g for the five alloy models lie between 0.63–0.87 eV (Table S3[†]), with near-quantitative agreement between the values for the 32-atom supercell enumeration and SQS models ($E_g = 0.63/0.64$ eV). The calculated bandgaps are all also lower than the value of $E_g = 1.06 \pm 0.15$ eV measured for SnS, but several of the models are below the $E_g = 0.86$ eV reported for SnSe.^{14,31} While codes to perform electronic band unfolding are available,^{85,86} they are not easily integrated into the present workflow, and we were therefore unable to obtain an averaged dispersion for the 32-atom supercell enumeration model. By analysing the “folded” dispersions of the four SQS models, it is

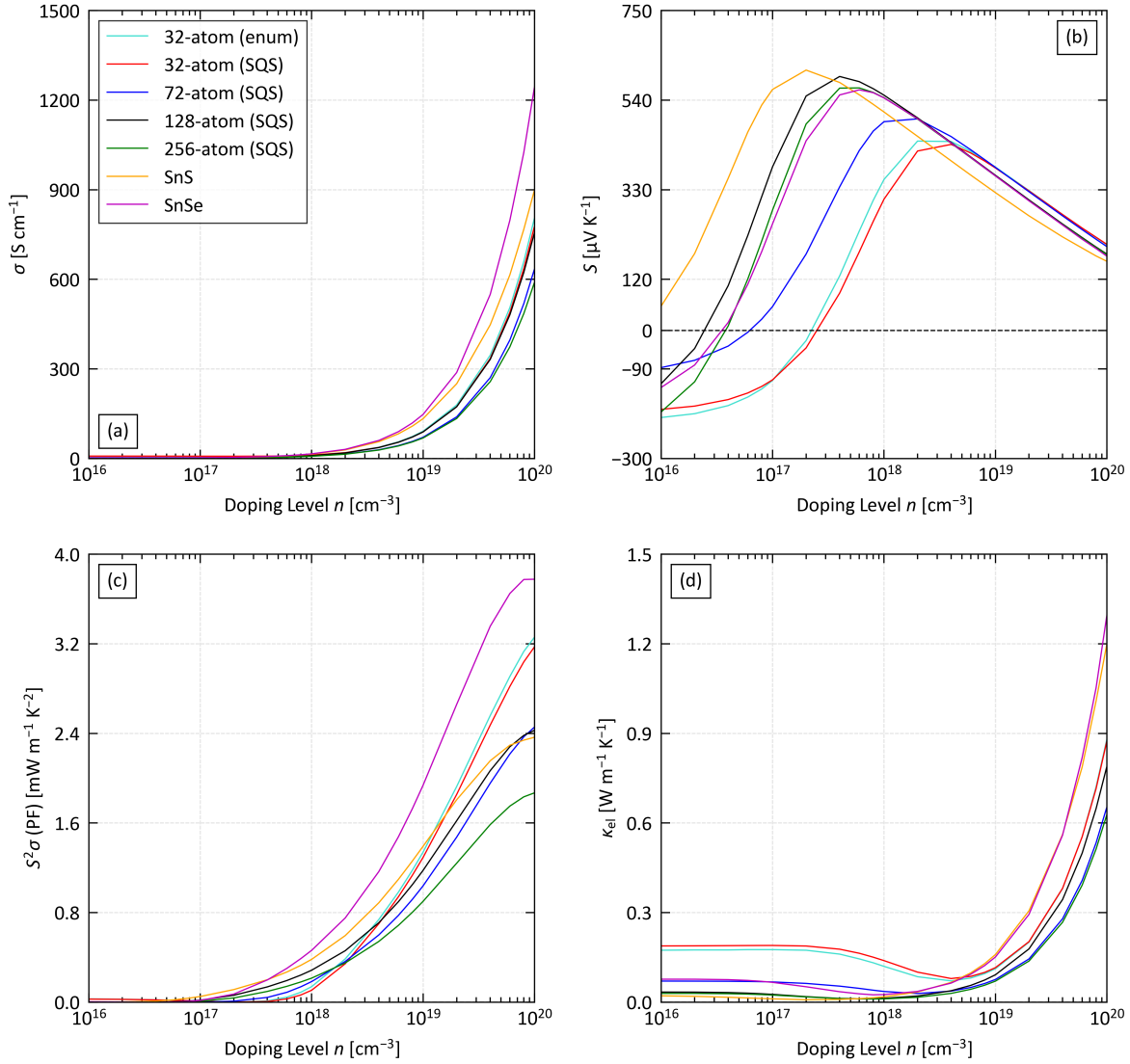


Fig. 5 Predicted scalar-average electrical transport properties of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work as a function of extrinsic hole carrier concentration ("doping level") n at a fixed $T = 700$ K: (a) electrical conductivity σ , (b) Seebeck coefficient S , (c) power factor $S^2\sigma$ (PF) and (d) electronic thermal conductivity κ_{el} . Data for the SnS and SnSe endpoints taken from our previous study are shown for comparison.⁴²

not possible to determine whether an apparent direct bandgap would remain so in the unfolded band structure, but it is possible to unambiguously assign an indirect gap. In this context, the folded band structures all indicate indirect bandgaps, matching the SnS and SnSe endpoints (Figs. S7-S10[†]). The fact that we predict some of the alloy models to possess lower E_g than SnSe is unusual and differs qualitatively from our previous work on the $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ alloys.⁵⁰ However, the E_g in Ref. 50 were not computed using a hybrid functional, and we would therefore expect the present calculations to be more accurate. In keeping with this, we recently applied the supercell-enumeration approach in the present work to model the electronic structure of $\text{Ge}(\text{S}_{0.5}\text{Se}_{0.5})$ and obtained very good agreement with experimental measurements.⁵¹

As outlined above, our procedure for computing the electrical properties involves solving the electron Boltzmann transport

equations using approximate models for the electron scattering rates (carrier lifetimes).^{38,39} In principle, this can be straightforwardly applied to both the supercell-enumeration and SQS approaches for modelling alloys since, unlike the third-order force constants $\Phi(3)$, calculating the required uniform band structure and physical properties remains tractable even for large supercells. One potential caveat is that the transport modelling performs Brillouin zone interpolation and thus requires sufficiently dense $\mathbf{k}$ -point sampling in the uniform band structure calculation. While this can be computationally expensive for large cells, our approach of computing the dense band structure with semi-local DFT and correcting the bandgap with a more accurate E_g computed using a hybrid functional on a smaller $\mathbf{k}$ -point mesh largely mitigates this, particularly when used in conjunction with our non-self-consistent approach.⁶⁶ A second, potentially more important consideration is that the transport distribution func-

tion in Eq. (15), like the κ_p in Eq. (10), depends on the electron group velocities $\mathbf{v}_{\mathbf{k}j}$, and the electrical properties could therefore show a similar systematic dependence on the supercell size. However, unlike for the κ_{latt} , we are not aware of a widely-available equivalent of the Wigner transport formulation for electrons. We explore this second point in detail in the following discussion.

Fig. 5 shows the predicted scalar-average p-type electrical conductivity σ , Seebeck coefficient S , power factor $S^2\sigma$ and electronic thermal conductivity κ_{el} of the five alloy models as a function of $n = 10^{16}$ – 10^{20} cm $^{-3}$ at a fixed $T = 700$ K chosen to be below the $Pnma \rightarrow Cmc$ phase transition of the endpoints ($T = 878$ and 807 K for SnS and SnSe respectively²⁷).

The electrical conductivity σ is conventionally expressed as:

$$\sigma = ne\mu = \frac{ne^2\tau_{\text{el}}}{m_{\sigma}^*} \quad (30)$$

where $\mu = e\tau/m_{\sigma}^*$ is the carrier mobility, τ_{el} is a weighted-average electron relaxation time analogous to the τ_{ph} in Eq. 8, and m_{σ}^* is the conductivity effective mass. The linear dependence on n leads to the characteristic steep rise in σ with n when plotted on a log-linear scale as in Fig. 5(a). Interestingly, at larger n all five alloy models are predicted to have lower conductivity than the SnS and SnSe endpoints. Two other noteworthy observations are also apparent. Firstly, the averaged σ calculated for the 32-atom supercell enumeration model is very similar to that of the (single) 32-atom SQS model, which suggests that, as for the E_g , the electrical properties are well modelled by a single configuration. Secondly, while the σ of the 32- and 128-atom models are very similar, the 72- and 256-atom models are predicted to have substantially lower conductivity. We tentatively attribute the lower σ of the 256-atom SQS model to the larger supercell, and that of the 72-atom structure to the fundamentally different Brillouin-zone folding in the “odd” supercell expansion used for this model (c.f. Table 2).

To explore the differences in conductivity between the five Sn(S $_{0.2}$ Se $_{0.8}$) models and the SnS/SnSe endpoints, we used the approach reported in Ref. 87 to estimate the m_{σ}^* and τ_{el} at a fixed $n = 4 \times 10^{19}$ cm $^{-3}$ and $T = 700$ K (Table S4 †). The carrier masses of the alloy models show an increase with supercell size, from $m_{\sigma}^* = 0.3/0.16 m_e$ for the two 32-atom models with p-/n-type doping to $m_{\sigma}^* = 0.4/0.2 m_e$ for the 256-atom model. The m_{σ}^* are inversely proportional to the band curvature $\partial^2\epsilon_{\mathbf{k}j}/\partial\mathbf{k}^2$, and the weak size dependence can be interpreted through the impact of disorder on the Brillouin zone folding (c.f. Fig. 4). In all cases the p-type m_{σ}^* are higher than those of SnS and SnSe, while the n-type m_{σ}^* of the smaller 32-atom models are between those of the two endpoints and closer to SnSe. The τ_{el} range from 8.47–10.57 ps for p-type doping and 9.56–11.6 ps for n-type doping, compared to 10.16/13.36 and 8.75/12.44 ps for p- and n-doped SnS/SnSe. Our models therefore suggest that alloying will likely result in higher carrier effective masses and shorter carrier lifetimes compared to SnSe, which accounts for the consistently lower predicted conductivity.

To further investigate the variation in the τ_{el} we compared the energy- and momentum-dependent electron scattering rates $\Gamma_{\mathbf{k}j}$

(i.e., the inverse of the lifetimes $\tau_{\mathbf{k}j}$) as a function of energy for the p- and n-type SQS models at $T = 700$ K and a typical $n = 4 \times 10^{19}$ cm $^{-3}$ ^{15,88–92} (Figs. S11–S14 †). As in our previous work on GeS/GeSe and SnS/SnSe,^{42–44} we find that the $\tau_{\mathbf{k}j}$ of the electronic states close to the Fermi energy are dominated by polar-optic phonon (POP) scattering, with ionised impurity (IMP) scattering becoming increasingly significant away from the valence- and conduction-band edges. This analysis therefore suggests that the alloys show similar carrier scattering physics to the endpoints, and the differences in τ_{el} are mainly due to small changes in the POP scattering rates.

The Seebeck coefficient can be related to the n through the Mott equation:

$$S = \frac{8\pi^2 k_B^2}{3qh^2} m_S^* \left(\frac{\pi}{3n} \right)^{2/3} \quad (31)$$

where $q \pm e$ for hole and electron carriers, respectively and m_S^* is the Seebeck effective mass. The m_S^* is related to the m_{σ}^* by the valley degeneracy N_V :

$$m_S^* = N_V^{2/3} m_{\sigma}^* \quad (32)$$

At low n the S of SnS, SnSe and of all five alloy models are predicted to be negative, indicative of intrinsic n-type behaviour that requires a minimum doping level of up to $\sim 2 \times 10^{17}$ cm $^{-1}$ to offset (Fig. 5(b)). The S initially rise with n and peak between $n \approx 2 \times 10^{17}$ – 5×10^{18} cm $^{-3}$ and then show the typical monotonic decrease at larger n . The minimum n required to obtain p-type conductivity, and the n where the peak S are obtained, show a strong correlation with the predicted bandgaps (Table S5 †). The shifts in the crossing points can be attributed to smaller bandgaps producing higher populations of thermally-generated charge carriers, and the shifts in the maxima further suggest that the differences in the E_g may arise predominantly from rigid shifts of the valence/conduction bands. At larger n the S show a weaker dependence on the supercell size, with the five models predicted to show a narrow range of $S = 248$ – 272 $\mu\text{V K}^{-1}$ at a typical “high” doping level of $n = 4 \times 10^{19}$ cm $^{-3}$.^{15,88–92} Interestingly, the S of the alloy models are very similar to the $S = 247$ $\mu\text{V K}^{-1}$ predicted for SnSe, and higher than the $S = 219$ $\mu\text{V K}^{-1}$ of SnS, from which we conclude that the alloys retain the favourable larger Seebeck coefficient of the selenide.

The power factors $S^2\sigma$ reflect the differences in both the S and the σ (Fig. 5(c)). This results in an interesting scenario where the alloys are predicted to show lower PFs than both endpoints at small n , but the PFs of the two 32-atom models cross that of SnS at $n \approx 2 \times 10^{19}$ cm $^{-1}$, and those of the 72- and 128-atom SQS models cross at higher $n \approx 5 \times 10^{19}$ cm $^{-3}$. Given that the optimum p-type doping level for the endpoints typically falls within this range of n ,^{15,88–92} our calculations are inconclusive on whether alloying should yield a PF between those of SnS and SnSe, or a PF comparable to SnS.

Finally, at large n the alloys and both the endpoints show degenerate semiconducting behaviour where the κ_{el} are proportional to the σ through the Wiedemann-Franz law:^{7,93}

$$\kappa_{\text{el}} = L\sigma T \quad (33)$$

where L is the Lorentz number and may in general depend on n and T .⁷ The κ_{el} therefore closely mirror the trends in the σ . At $n = 4 \times 10^{19} \text{ cm}^{-3}$ we predict $\kappa_{\text{el}} = 0.56 \text{ W m}^{-1} \text{ K}^{-1}$ for SnSe, $0.34\text{--}0.38 \text{ W m}^{-1} \text{ K}^{-1}$ for the 32- and 128-atom alloy models, $0.27 \text{ W m}^{-1} \text{ K}^{-1}$ for the 256-atom model, and $0.56 \text{ W m}^{-1} \text{ K}^{-1}$ for SnS. This means the κ_{el} of the alloy models are predicted to be between 57–68% of the predicted κ_{latt} at $T = 700 \text{ K}$ (c.f. Fig. 3). It is also noteworthy that the κ_{el} show an unusual fall around the n where the S change sign, and that the two 32-atom models are predicted to show a significant $\kappa_{\text{el}} \approx 0.18 \text{ W m}^{-1} \text{ K}^{-1}$ at low n due to the narrow bandgaps and high thermally-generated electron concentrations.

As for the κ_{latt} , there are a handful of experimental measurements of the electronic properties of $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$,^{33–35} to which it is more representative to compute and compare properties along the out-of-plane (parallel) and in-plane (perpendicular) directions (c.f. Eqs 27/28). Ref. 33 reports room-temperature values of $S^{\parallel} = 312 \text{ } \mu\text{V K}^{-1}$ and $\sigma^{\parallel} \approx 10 \text{ S cm}^{-1}$ for p-type $(\text{Ag}_{0.01}\text{Sn}_{0.99})(\text{S}_{0.20}\text{Se}_{0.80})$ with $n = 6.22 \times 10^{18} \text{ cm}^{-3}$. At a comparable $n = 6 \times 10^{18} \text{ cm}^{-3}$ we predict values of $S^{\parallel} = 281\text{--}316 \text{ } \mu\text{V K}^{-1}$ and $\sigma^{\parallel} = 41\text{--}66 \text{ S cm}^{-1}$ across our five alloy models. Ref. 34 reports a p-type $\sigma^{\parallel} \approx 0.5 \text{ S cm}^{-1}$ at 323 K for $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ with a lower $n \approx 2.5 \times 10^{17} \text{ cm}^{-3}$, and we again predict comparable $\sigma^{\parallel} = 1.4\text{--}2.3 \text{ S cm}^{-1}$ at 320 K and $n = 2 \times 10^{17} \text{ cm}^{-3}$. The $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ samples prepared in Ref. 35 have a similar $n = 2.1 \times 10^{17} \text{ cm}^{-3}$ and measured p-type $S^{\perp} = 265/S^{\parallel} = 279 \text{ } \mu\text{V K}^{-1}$ and $\sigma^{\perp} = 2.8 \times 10^{-3}/\sigma^{\parallel} = 1.53 \times 10^{-3}$ at 323 K .³⁵ This is a more interesting point of comparison to the calculations because the low n falls in the region of carrier concentrations where the predicted S of the alloy models vary significantly (Fig. 5(b)). The closest $S^{\perp} = 567$ and $S^{\parallel} = 575 \text{ } \mu\text{V K}^{-1}$ are obtained for the 128-atom SQS model, but clearly do not match the experiments. Moreover, the calculated $\sigma^{\perp} = 6.06\text{--}8.69$ and $\sigma^{\parallel} = 1.42\text{--}2.27 \text{ S cm}^{-1}$ predicted by the alloy models are orders of magnitude larger than the measurements. These discrepancies are likely attributable in part to the high sensitivity of the Seebeck coefficients to the bandgap at low n . However, given the more reasonable match between the calculations and the other sets of experiments, the discrepancy could also be indicative of a high level of defects in the experimental samples in Ref. 35, which could modulate both the S and σ .^{94,95} We are therefore reluctant to read too much into this comparison.

While we have followed experiments and focused our discussion on the more usual p-type doping,^{15,88,89} both SnS and SnSe can also be n-type doped,^{90–92} and we therefore performed calculations for the n-type alloys (Fig. S15[†]). Notably, the n-type σ of the alloys show a clearer systematic trend in supercell size. The two 32-atom models have very similar and slightly larger σ than the 128-atom SQS models, with the predicted conductivities of all three models falling between the SnS and SnSe endpoints, whereas the larger 256-atom SQS model is predicted to have a much lower σ comparable to SnS. As for the p-type S , the n-type Seebeck coefficients are predicted to reach peak (minimum) values between $n \approx 10^{17}\text{--}10^{18} \text{ cm}^{-3}$, and the positions of the peaks again correlate to the bandgaps. At large n , the S of the 32-, 128- and 256-atom alloy models are very similar, spanning a narrow

range of $|S| = 235\text{--}255 \text{ } \mu\text{V K}^{-1}$ at $n = 4 \times 10^{19} \text{ cm}^{-3}$, which is very similar to the n-type $|S| = 230$ and 227 predicted for SnS and SnSe. The 72-atom alloy model appears to be something of an outlier, with a lower σ and larger (more negative) S . This potentially provides further evidence for different Brillouin zone folding in the supercell expansion used for this model. Leaving this model aside, the clear contrast between the the strong and weak dependence of the σ and S on the supercell size in the n-type alloys is noteworthy because the latter are calculated from the ratio of two moments of the transport distribution functions, and any systematic size effects in the $\mathbf{v}_{\mathbf{k}j}$ should largely cancel (c.f. Eqs (15), (16), (19) and (20)). In keeping with this, the n-type PFs strongly reflect the behaviour of the σ , with the notable exception that the PF of the SnS endpoint appears to plateau at large $n \rightarrow 10^{20} \text{ cm}^{-3}$. Finally, the κ_{el} closely follow the σ , as for the p-type systems, with the result that the predicted κ_{el} are $1.2\text{--}1.6\times$ the κ_{latt} at $n = 4 \times 10^{19} \text{ cm}^{-3}$ and $T = 700 \text{ K}$. When taken with the weak size dependence of the S , this is of note because it means that any systematic supercell size dependencies in the σ and κ_{el} should partially cancel in the zT . We return to this point in the following subsection.

For completeness, we also compare the predicted σ , S , $S^2\sigma$ and κ_{el} of the five alloy models and the SnS and SnSe endpoints as a function of temperature at a fixed $n = 4 \times 10^{19} \text{ cm}^{-3}$ in Figs. S16 and S17.[†] At this high doping level the σ show a metallic-like decrease with temperature, confirming the degenerate semi-conducting behaviour inferred from the κ_{el} . On the other hand, the absolute S increase with temperature (i.e., the p- and n-type S become more positive/negative), and the PFs show a shallow temperature dependence with a broad peak from $300\text{--}600 \text{ K}$. For the electronic thermal conductivity the sharp drop in the σ is partially counteracted by the temperature term in Eq. (33), and the κ_{el} show a small, close to linear, decrease of $22\text{--}57\%$ between $300\text{--}900 \text{ K}$.

Finally, and again for completeness, we also compare the predicted in-plane and out-of-plane electrical transport properties of the p- and n-type 32-atom supercell enumeration and 128-atom SQS models, both as a function of n at fixed T and *vice versa*, to the corresponding scalar averages in Figs. S18–S21.[†] As found in our previous studies on SnS and SnSe,^{42,43} the p- and n-type alloys are both predicted to have relatively isotropic S , but higher σ , $S^2\sigma$ and κ_{el} along the in-plane and out-of-plane directions respectively. This means that for n-type doping the “easy” axis for electronic transport aligns with the “hard” axis for thermal transport, which is beneficial for a higher zT .⁴³ We note that our qualitative prediction of a higher σ and PF along the in-plane ($\perp$) direction in the p-type alloys is consistent with the experimental measurements in Ref. 35, despite the fact that our predicted values are far from quantitative.

3.3 Thermoelectric figure of merit

By combining the scalar-average electrical properties and lattice thermal conductivity, we obtain a predicted scalar-average figure of merit zT for each of the five alloy models as a function of doping level and temperature (Figs. 6 and S22[†]). From these, we

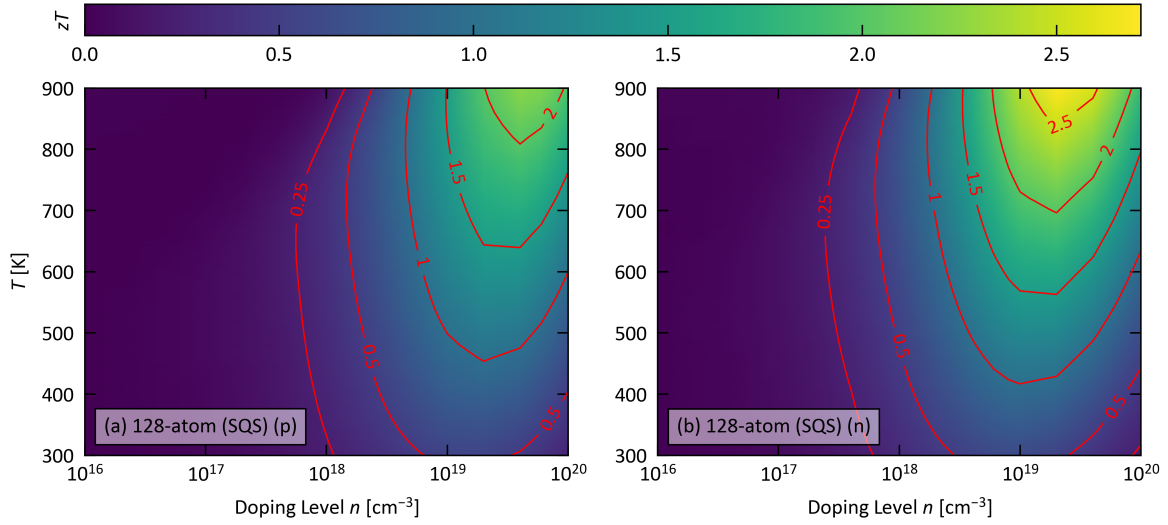


Fig. 6 Predicted thermoelectric figure of merit zT of p- and n-type $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ modelled using the 128-atom special quasi-random structure (SQS) model, as a function of extrinsic electron carrier concentration ("doping level") n and temperature T . Values of $zT = 0.25, 0.5, 1, 1.5, 2$, and 2.5 are shown by contour lines.

then determine the maximum zT (zT_{max}) of these alloy models at temperatures up to the reported phase transition temperatures of the endpoints,²⁷ which are collected together with the constituent properties in Table 3. We also include data for SnS and SnSe from our previous work,^{42,43} including the wave-like contribution to the lattice thermal conductivity for consistency with the calculations on the alloys.⁴⁴

All five alloy models are predicted to achieve the highest zT values at doping levels of $2 - 4 \times 10^{19} \text{ cm}^{-3}$ and the highest $T = 800 \text{ K}$ we tested, which matches SnS and SnSe.^{42,43} The higher n are required to optimise the σ and power factor $S^2\sigma$, while higher T minimises the thermal conductivity and in particular the κ_{latt} (c.f. Figs. 3 and 5). These doping levels are readily obtained in experiments on SnS and SnSe,^{15,88,91,92,96,97} and should therefore also be attainable in $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ alloys.

The predicted zT of the alloy models shows a clear size dependence, with the two 32-atom models yielding comparable and relatively large zT and the 256-atom SQS model yielding a relatively small zT . Across the five alloy models the p- and n-type zT vary by a factor of 1.3 and $1.1\times$, respectively, which is predominantly due to differences in the electrical properties. The p- and n-type PFs vary by $1.6/1.3\times$, but these are partially offset by similar $1.5/1.3\times$ variation in the κ_{el} , which accounts for 32–52% of the predicted (total) thermal conductivity. In contrast, the κ_{latt} shows a systematic but much smaller variation with system size, falling from 0.51/0.52 for the two 32-atom models to 0.48 for the 256-atom model. Given that we are confident our approach correctly accounts for the crossover from dominant particle-like to wave-like transport with increasing disorder, this small reduction may be a real consequence of the better representation of the disorder in the larger cells. As discussed in the previous section, we attribute the systematically lower σ of the 256-atom SQS model

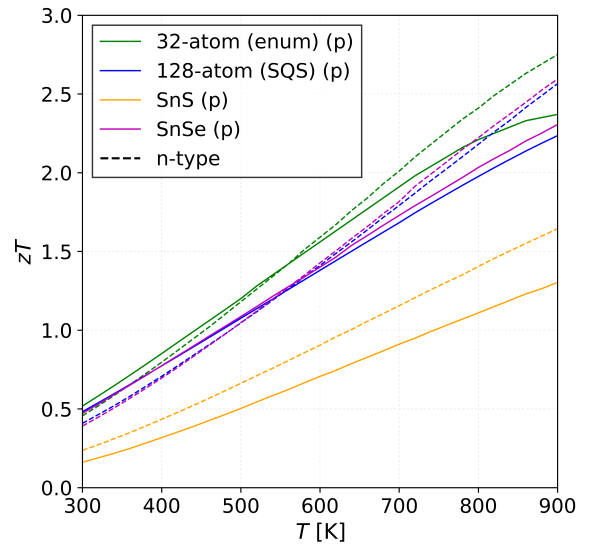


Fig. 7 Predicted thermoelectric figure of merit zT of p- and n-type SnS, SnSe and $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ alloys as a function of temperature at a fixed extrinsic electron carrier concentration ("doping level") $n = 4 \times 10^{19} \text{ cm}^{-3}$. The data for *Pnma* SnS and SnSe are taken from our previous work^{42,43} and the κ_{latt} include the contribution from intraband tunnelling.⁴⁴

to an artifact of the system size. Bearing in mind the $2\times$ larger n at which the n-type zT_{max} is obtained, the smaller 72-atom model is also predicted to show an anomalously low electrical conductivity, which, again as discussed above, we attribute to the different Brillouin-zone folding in this supercell expansion. Based on these observations, and the representation of the disorder as inferred from the unfolded phonon band structures in Fig. 2, we consider the 32-atom supercell enumeration and 128-atom SQS models to

Table 3 Predicted maximum scalar-average thermoelectric figure of merit $zT_{\max}$ of the five $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ models examined in this work, with both p- and n-type doping. For each model we list the extrinsic carrier concentration (“doping level”) n and temperature at which the $zT_{\max}$ is attained, together with the associated electrical conductivity σ , Seebeck coefficient S , power factor $S^2\sigma$ (PF), and electronic, lattice and total thermal conductivity κ_{el} , κ_{latt} and κ . The κ_{latt} of all five alloy models are computed using estimated average three-phonon interaction strengths $\bar{P}$. Data for $Pnma$ SnS and SnSe taken from our previous work^{42,43} are shown for comparison, and the κ_{latt} include the contribution from intraband tunnelling.⁴⁴

		n	T	zT	σ	S	PF $S^2\sigma$	κ [$\text{W m}^{-1} \text{K}^{-1}$]		
		[cm^{-3}]	[K]		[S cm^{-1}]	[$\mu\text{V K}^{-1}$]	[$\text{mW m}^{-1} \text{K}^{-2}$]	κ_{el}	κ_{latt}	κ
p-type	32-atom (enum)	4×10^{19}	800	2.21	285	287	2.36	0.34	0.52	0.86
	32-atom (SQS)	4×10^{19}	800	2.15	277	288	2.29	0.34	0.51	0.85
	72-atom (SQS)	4×10^{19}	800	1.95	223	284	1.79	0.24	0.50	0.74
	128-atom (SQS)	4×10^{19}	800	1.99	274	266	1.95	0.30	0.49	0.79
	256-atom (SQS)	4×10^{19}	800	1.67	210	265	1.47	0.23	0.48	0.71
	SnS ^{42,43,52}	4×10^{19}	880	1.26	305	255	1.98	0.46	0.92	1.38
	SnSe ^{42,43,52}	4×10^{19}	800	2.03	448	262	3.07	0.49	0.72	1.21
n-type	32-atom (enum)	2×10^{19}	800	2.56	304	-314	3.00	0.42	0.52	0.94
	32-atom (SQS)	2×10^{19}	800	2.53	299	-315	2.96	0.42	0.51	0.93
	72-atom (SQS)	4×10^{19}	800	2.22	389	-272	2.88	0.54	0.50	1.04
	128-atom (SQS)	2×10^{19}	800	2.40	282	-310	2.70	0.41	0.49	0.90
	256-atom (SQS)	2×10^{19}	800	2.26	237	-308	2.25	0.32	0.48	0.80
	SnS ^{42,43,52}	4×10^{19}	880	1.60	386	-262	2.65	0.54	0.92	1.46
	SnSe ^{42,43,52}	2×10^{19}	800	2.38	400	-299	3.58	0.48	0.72	1.20

strike the best balance of an adequate description of the disorder and minimising supercell-size artifacts in the predicted electrical properties, and so we focus our discussion on these models. We note, however, that the single 32-atom SQS model does seem to be a good proxy for the averaged electronic and thermal transport of the supercell enumeration model (Tables S6/S7[†]).

Both models predict that alloys with composition close to $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ should show significantly higher zT than SnS and comparable or higher zT than SnSe. With p-type doping, alloying reduces the σ but also the κ_{el} , which, in combination with the reduced κ_{latt} , leads to a predicted reduction in the total high-temperature κ of around 40% compared to SnS and 30% compared to SnSe. The 128-atom model is predicted to have a similar PF to SnS, whereas the 32-atom model is predicted to have a 20% higher $S^2\sigma$ that lies between those of SnS and SnSe, which leads to a higher predicted zT . n-type SnS and SnSe are predicted to show superior performance to their p-type counterparts due to a higher σ (for a given n), a larger absolute $|S|$, and a higher PF.^{42,43,52} Compared to endpoints, the two alloy models are predicted to show lower σ , comparable or larger $|S|$, and lower total κ , which again result in comparable or slightly higher zT than SnSe.

Due to the strong anisotropy in the electronic and thermal transport, the zT of both endpoints and the alloy models also show significant anisotropy. We therefore calculated and compared the in-plane ($\perp$) and out-of-plane ($\parallel$) zT and $zT_{\max}$ with the scalar averages (Fig. S23/S24 and Table S8[†]). In the p-type systems, the in-plane and out-of-plane directions represent the easy and hard axes for electrical and thermal transport, respectively, and the highest zT are obtained along the in-plane direction. In the n-type systems, the out-of-plane direction is the easy axis for electrical transport, and the highest zT are obtained along the out-of-plane direction. Under the conditions listed in Table 3, the 32-atom supercell enumeration and 128-atom SQS models are predicted to have in-plane p-type $zT^{\perp} = 2.28/2.18$, which are slightly larger than the p-type $zT^{\perp} = 2.09$ for SnSe.⁴² Similarly,

the two models are predicted to have out-of-plane n-type $zT^{\parallel} = 4.61/4.35$, compared to $zT^{\parallel} = 4.25$ for SnSe.⁴³

While most experimental work on the Sn chalcogenides has focused on the “headline” high-temperature zT , the low-temperature zT are more relevant in many heat-recovery scenarios.^{6,98} Fig. 7 compares the predicted zT as a function of temperature of SnS, SnSe and the 32-atom supercell enumeration and 128-atom SQS models of $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ at a common $n = 4 \times 10^{19} \text{ cm}^{-3}$. This analysis shows that the alloys have zT competitive with or higher than SnSe at high temperature, but we also predict modest increases of 10 and 17% in the p- and n-type zT at room temperature, and we predict that the p-type 32-atom supercell enumeration model would attain a benchmark $zT = 1$ at 450 K. We attribute the improved low-temperature performance to the significant reduction in the low- T κ_{latt} , which is partially due to the shallower T dependence of the κ_{w} that dominates in more disordered systems.⁴⁴ Our calculations therefore highlight alloying as a potential route to improving the low-temperature zT .

The measurements on $(\text{Ag}_{0.01}\text{Sn}_{0.99})(\text{Se}_{0.80}\text{S}_{0.20})$ in Ref. 33 report a p-type $zT^{\parallel} = 1.67$ at $n = 6.22 \times 10^{18} \text{ cm}^{-3}$ and $T = 823 \text{ K}$. This is significantly larger than our calculated $zT^{\parallel} = 0.17/0.53$ for the 32-atom supercell enumeration and 128-atom SQS models at a comparable $n = 6 \times 10^{18} \text{ cm}^{-3}$ and $T = 820 \text{ K}$. This appears to be largely due to our higher predicted $\kappa^{\parallel} = 0.33/0.36 \text{ W m}^{-1} \text{K}^{-1}$, which is more than a factor of two larger the $\kappa^{\parallel} = 0.15 \text{ W m}^{-1} \text{K}^{-1}$ measured in the experiments.³³ We note, however, that we predict comparable $zT = 1.14/1.39$ along the perpendicular direction. Ref. 34 measured values of $zT^{\parallel} = 0.82$ and $zT^{\perp} = 0.62$ for $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ at $n \approx 2.5 \times 10^{17} \text{ cm}^{-3}$ and $T = 820 \text{ K}$, which are approximately an order of magnitude larger than our predicted $zT^{\parallel} = 0.70/0.09$ and $zT^{\perp} = 0.01/0.08$ at comparable $n = 2 \times 10^{17} \text{ cm}^{-3}$ and $T = 820 \text{ K}$. In this case, the discrepancy arises from a combination of an 18–50% larger predicted κ and a 75–76% lower predicted σ . Our calculations using the 128-atom SQS model also underestimate the similar p-type $zT^{\parallel} = 0.64$ mea-

sured for the $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ samples in Ref. 35 ($zT^{\parallel} = 0.70/0.09$), but in this case because the predicted $S^2\sigma = 0.05 \text{ mW m}^{-1} \text{ K}^{-2}$ is approximately $6 \times$ lower than the measured PF. We note, however, that the $n \approx 2 \times 10^{17} \text{ cm}^{-3}$ of the samples in Ref. 34/ 35 are in the region where the Seebeck coefficients are highly sensitive to the electronic bandgap – indeed, our 32-atom supercell enumeration model is predicted to show intrinsic n-type behaviour at this n (c.f. Fig. 5) – and thus where our predictions are likely to be most variable. With this in mind, we would expect our predictions to be most accurate at the larger n that optimise the zT , although the comparison to Ref. 33 suggests it might be prudent to explore the effect of crystallite size on the κ_{latt} .⁹⁹

4 Discussion and conclusions

In this study, we have implemented and benchmarked a first-principles modelling workflow for predicting the electronic and thermal transport properties and thermoelectric figure of merit zT of alloys.

Using the characteristic “smearing” of the high-frequency chalcogen modes in the unfolded phonon dispersion as a reference, we find that a (single) 128-atom special quasi-random structure (SQS) model of $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ gives comparable results to a thermodynamic average over a complete set of 23 symmetry-independent configurations in a 32-atom supercell. While on this basis we would recommend a “ballpark” size of ~ 100 atoms as a good balance between computational cost and capturing the range of local atomic environments in the alloy, we also found that, for this system, a single 32-atom SQS model provides very similar predictions to those obtained by enumerating and averaging over the full set of configurations.

The approach of using averaged three-phonon interaction strengths interpolated from the endpoint structures⁵² appears to provide a very good estimate of the lattice thermal conductivity obtained by explicit computation of the third-order force constants. Crucially, this method also allows both the particle- and wave-like contributions to the κ_{latt} to be evaluated, which captures the progressive change in character to amorphous-like thermal transport with the reduction of (imposed, artificial) periodicity in larger supercells.⁵³

The approach of using semi-classical Boltzmann transport theory with approximate models for the electron relaxation times^{38,39} can be straightforwardly applied to alloys, but is based on solving the electronic Boltzmann transport equations and assumes particle-like transport. This appears to result in a systematic size dependence where models based on large supercells predict anomalously low electrical conductivity. At the typical high doping levels that optimise the zT , this is likely in most cases to be partially mitigated by the similar impact on the electronic thermal conductivity, although we would recommend explicit tests for size dependencies by, for example, comparing against calculations on small supercells.

Overall, the present results suggest that the modelling approach presented here, with appropriate supercell sizes, is capable of realistic qualitative predictions of the effects of alloying on the thermoelectric performance. All our models predict the expected reduction in the κ_{latt} due to the alloying, as well as

the reduced σ we would intuitively expect if alloying has similar impacts on both the electronic and phonon band dispersions. However, predictions of the Seebeck coefficient are made somewhat challenging by the narrow bandgap, which means that small changes in the predicted E_g across different alloy models translate into large variations in the S at low doping levels. Despite this issue, our models predict that the highest zT are obtained under comparable conditions to the endpoints, and suggest a modest enhancement in the zT that qualitatively matches experimental findings. Another important qualitative finding is that the shift from predominantly particle-like κ_{latt} in the endpoints to wave-like transport leads to a different character of temperature dependence that favours a reduced low- T thermal conductivity and higher low- T zT .⁴⁴

On the other hand, the small number of experimental studies makes it somewhat difficult to assess the quantitative accuracy of our predictions. In particular, many of the $\text{Sn}(\text{S}_{1-x}\text{Se}_x)$ thermoelectric materials reported to date appear to have been prepared with relatively low (i.e., unoptimised) n , and the considerable variation in the measured properties suggests a strong potential impact of different preparation conditions. With this in mind, the fact that the base workflow has been shown to provide quantitative predictions of the zT of SnS/SnSe and related materials that have been more extensively characterised,^{42,43,99,100} and that the predictions we obtain for $\text{Sn}(\text{S}_{0.2}\text{Se}_{0.8})$ are consistent with the typical behaviour expected for alloys, we are confident that our approach should be able to make quantitative predictions of the zT of bulk alloys under optimal conditions.

Future development of the methodology should focus on improving the description of the electrical transport to address the dependence on supercell size by accounting for both particle- and wave-like electrical transport in an analogous manner to the Wigner formulation of the κ_{latt} .⁶¹ One possible approach is the non-equilibrium Green’s function method, where a bias is applied to the model through virtual contacts (e.g., Ref. 101). Another option may be to calculate the σ from the DC ($E = 0$) component of the optical conductivity, and the Seebeck coefficient and κ_{el} from phenomenological relationships such as the Mott relation and the Wiedemann-Franz law.^{93,102} The latter has the advantage of simplicity, but with present implementations it would likely be difficult to include per-state electronic linewidths, and the approach would only be applicable under certain conditions (e.g., the degenerate limit for the Wiedemann-Franz law).

Author Contributions

Conceptualisation - JMS; methodology - all authors; data curation - all authors; formal analysis and investigation - all authors; writing (original draft) - MZ and JMF; writing (review and editing) - all authors; funding acquisition - JMS. All authors have read and agreed to the published version of the paper.

Conflicts of interest

There are no conflicts to declare.

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Data availability statement

Raw data from the calculations performed in this study will be made available to download free of charge following acceptance from an online repository at DOI: 10.17632/tsdyh5kk46.

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